

```
TC
File Edit Run Compile Project Options Debug Break/watch
Line 15 Col 2 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int n,r,c;
clrscr();
printf("Enter no of rows "); scanf("%d",&n);
for(r=1;r<=n;r++)
{
for(c=1;c<=n;c++)
{
if(c<=n-r)printf("%3d",1); else printf("%3d",r);
}
printf("\n");
}
getch();
}
```

Enter no of rows 4

1	1	1	1
1	1	2	2
1	3	3	3
4	4	4	4

```

Enter no of rows 10
1 1 1 1 1 1 1 1 1 1
1 1 1 1 1 1 1 1 2 2
1 1 1 1 1 1 1 3 3 3
1 1 1 1 1 1 4 4 4 4
1 1 1 1 1 5 5 5 5 5
1 1 1 1 6 6 6 6 6 6
1 1 1 7 7 7 7 7 7 7
1 1 8 8 8 8 8 8 8 8
1 9 9 9 9 9 9 9 9 9
10 10 10 10 10 10 10 10 10 10

```

```

for(r=1;r<=4;r++)
{
for( c=1; c<=4; c++)
{
if(c<=n-r) p(1);else p(r);
}
p("\n");
}

```

n	r	1	r
4	-	1	3 ✓
4	-	2	2 ✓
4	-	3	1 ✓
4	-	4	0 ✓

1	1	1	1	-	1
1	1	2	2	-	2
1	3	3	3	-	3
4	4	4	4	-	4

c	$<=$	$n-r$	
1 2 3 4	$4-1=3$	1 1 1 1	
1 2 3 4	$4-2=2$	1 1 2 2	
1 2 3 4	$4-3=1$	1 3 3 3	
1 2 3 4	$4-4=0$	4 4 4 4	

```
TC
File Edit Run Compile Project Options Debug Break/watch
Line 12 Col 49 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int nr, nc, r, c, a;
clrscr();
printf("Enter no of rows and columns "); scanf("%d %d",&nr, &nc);
for(r=1; r<=nr; r++)
{
a = nc;
for(c=1; c<=nc; c++)
{
if(r%2==0)printf("%3d",a--); else printf("%3d",c);
}
printf("\n");
}
getch();
}
```

Enter no of rows and columns 4 4

```
1 2 3 4
4 3 2 1
1 2 3 4
4 3 2 1
```

```

TC
Enter no of rows and columns 5 9
1 2 3 4 5 6 7 8 9
9 8 7 6 5 4 3 2 1
1 2 3 4 5 6 7 8 9
9 8 7 6 5 4 3 2 1
1 2 3 4 5 6 7 8 9

```

```

for(r=1;r<=nr; r++)
{ a=nc;
  for ( c=1; c<=nc; c++)
  {
    if( r%2==0) p(a--);
    else p(c);
  }
}

```

$\frac{nr}{4}$ $\frac{nc}{4} \rightarrow \frac{a--}{4 \ 3 \ 2 \ 1}$

$\begin{array}{cccc} \checkmark & \checkmark & \checkmark & \checkmark \\ 1 & 2 & 3 & 4 \\ \checkmark & \checkmark & \checkmark & \checkmark \\ 4 & 3 & 2 & 1 \\ \checkmark & \checkmark & \checkmark & \checkmark \\ 1 & 2 & 3 & 4 \\ \checkmark & \checkmark & \checkmark & \checkmark \\ 4 & 3 & 2 & 1 \end{array}$

```
TC
File Edit Run Compile Project Options Debug Break/watch
Line 12 Col 51 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int nr, nc, r, c, a=1;
clrscr();
printf("Enter no of rows and columns "); scanf("%d %d",&nr, &nc);
for(r=1; r<=nr; r++)
{
for(c=1; c<=nc; c++)
{
if(r%2==0)printf("%3d",--a); else printf("%3d",a++);
}
printf("\n");
}
getch();
}
```

Enter no of rows and columns 8 20

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1

```

a=1
for(r=1;r<=nr; r++)
{
    for ( c=1; c<=nc; c++)
    {
        if( r%2==0) p(--a);
        else p( a++);
    }
}

```

$$\frac{nr}{4} \quad \frac{nc}{4} \quad \frac{a}{1}$$

```

      ✓✓✓✓
      1 2 3 4 — 1
      ✓✓✓✓
      4 3 2 1 — 3
      ✓✓✓✓
      1 2 3 4 — 3
      4 3 2 1

```

```
TC
File Edit Run Compile Project Options Debug Break/watch
Line 12 Col 16 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int nr, nc,r,c, a;
clrscr();
printf("Enter no of rows and columns "); scanf("%d %d",&nr, &nc);
for(r=1;r<=nr;r++)
{
for(a=r, c=1;c<=nc;c++)
{
printf("%3d",a++);
}
printf("\n");
}
getch();
}
```

Enter no of rows and columns 4 5

1	2	3	4	5
2	3	4	5	6
3	4	5	6	7
4	5	6	7	8

```

for(r=1;r<=nr; r++)
{
  a=r;
  for ( c=1; c<=nc; c++)
  {
    p( a++);
  }
}

```

$\frac{nr}{4}$	$\frac{nc}{4}$	$\frac{r}{1}$	$\rightarrow$	$\frac{a}{1\ 2\ 3\ 4}$	
		2		2 3 4 5	1 2 3 4 ✓
		3		3 4 5 6	2 3 4 5 ✓
					3 4 5 6
					4 5 6 7
					5 6 7 8


```
TC
File Edit Run Compile Project Options Debug Break/watch
Line 10 Col 5 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int nr, nc,r,c;
clrscr();
printf("Enter no of rows and columns "); scanf("%d %d",&nr, &nc);
for(r=1;r<=nr;r++)
{
for(c=1;c<=nc;c++)
{
printf("%3d",r+c-1);
}
printf("\n");
}
getch();
}
```

Enter no of rows and columns 10 5

1	2	3	4	5
2	3	4	5	6
3	4	5	6	7
4	5	6	7	8
5	6	7	8	9
6	7	8	9	10
7	8	9	10	11
8	9	10	11	12
9	10	11	12	13
10	11	12	13	14

```

for(r=1;r<=nr; r++)
{
  for ( c=1; c<=nc; c++)
  {
    p(a++);
  }
}

```

$\frac{nr}{4}$	$\frac{nc}{4}$	$\frac{r+c-1}{1 \quad 2 \quad 3 \quad 4}$
		2 1 2 3 4
		3

✓✓✓✓✓
1 2 3 4 ✓
<u>2 3 4 5</u> ✓
3 4 5 6
4 5 6 7
5 6 7 8

```
TC
File Edit Run Compile Project Options Debug Break/watch
Line 12 Col 18 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int nr, nc,r,c;
clrscr();
printf("Enter no of rows and columns "); scanf("%d %d",&nr, &nc);
for(r=1;r<=nr;r++)
{
for(c=1;c<=nc;c++)
{
printf("%2c",c+64);
}
printf("\n");
}
getch();
}
```

Enter no of rows and columns 10 26

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z

```
TC
File Edit Run Compile Project Options Debug Break/watch
Line 12 Col 18 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int nr, nc,r,c;
clrscr();
printf("Enter no of rows and columns "); scanf("%d %d",&nr, &nc);
for(r=1;r<=nr;r++)
{
for(c=1;c<=nc;c++)
{
printf("%2c",c+96);
}
printf("\n");
}
getch();
}
```

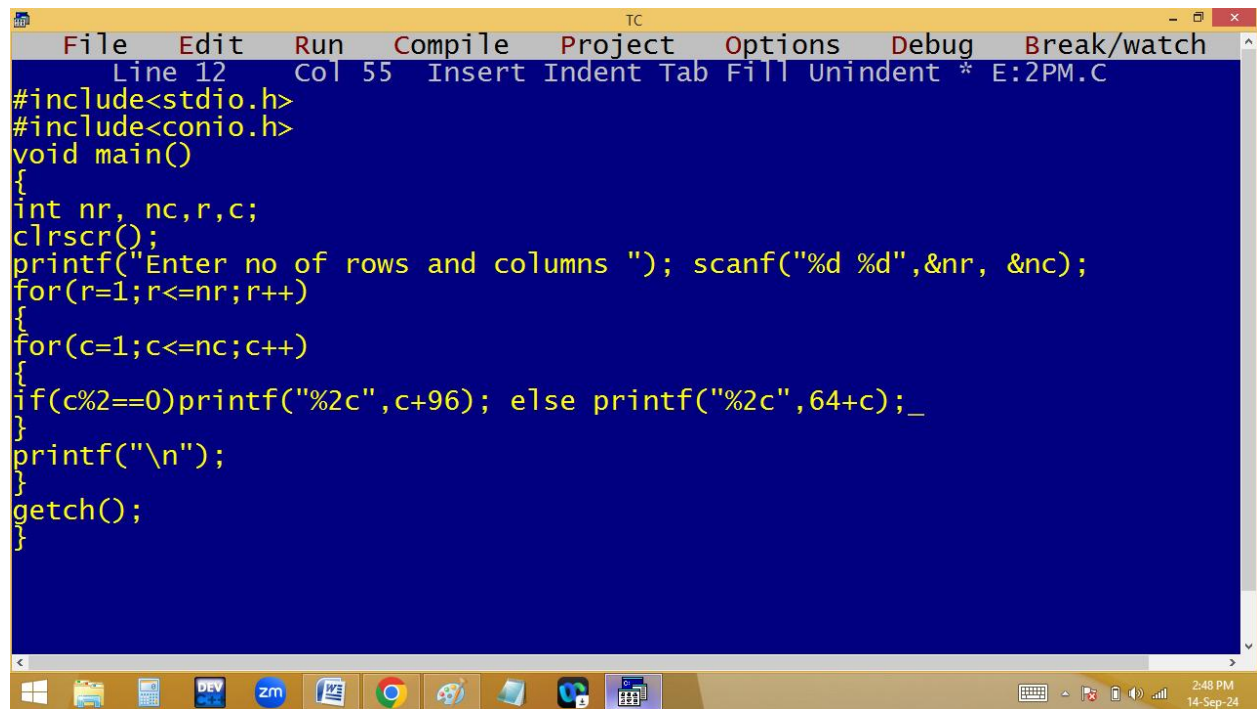
Enter no of rows and columns 10 26

a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z
a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z
a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z
a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z
a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z
a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z
a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z
a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z
a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z
a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z

```
TC
File Edit Run Compile Project Options Debug Break/watch
Line 12 Col 15 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int nr, nc,r,c;
clrscr();
printf("Enter no of rows and columns "); scanf("%d %d",&nr, &nc);
for(r=1;r<=nr;r++)
{
for(c=1;c<=nc;c++)
{
printf("%2c",r+96);
}
printf("\n");
}
getch();
}
```

Enter no of rows and columns 10 10

```
a a a a a a a a a a
b b b b b b b b b b
c c c c c c c c c c
d d d d d d d d d d
e e e e e e e e e e
f f f f f f f f f f
g g g g g g g g g g
h h h h h h h h h h
i i i i i i i i i i
j j j j j j j j j j
```



The image shows a screenshot of a Turbo C++ (TC) IDE window. The title bar reads "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", "Debug", and "Break/watch". The status bar at the top indicates "Line 12", "Col 55", and "Insert Indent Tab Fill Unindent * E:2PM.C". The main editing area has a dark blue background with yellow text. The code is as follows:

```
#include<stdio.h>
#include<conio.h>
void main()
{
    int nr, nc,r,c;
    clrscr();
    printf("Enter no of rows and columns "); scanf("%d %d",&nr, &nc);
    for(r=1;r<=nr;r++)
    {
        for(c=1;c<=nc;c++)
        {
            if(c%2==0)printf("%2c",c+96); else printf("%2c",64+c);_
        }
        printf("\n");
    }
    getch();
}
```

The Windows taskbar is visible at the bottom, showing icons for Windows, File Explorer, a calculator, DEV C++, ZOOM, Word, Chrome, Paint, a folder, and a game. The system tray on the right shows the time as 2:48 PM and the date as 14-Sep-24.


```
TC
File Edit Run Compile Project Options Debug Break/watch
Line 14 Col 38 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int nr, nc,r,c; char L='a', U='A';
clrscr();
printf("Enter no of rows and columns "); scanf("%d %d",&nr, &nc);
for(r=1;r<=nr;r++)
{
for(c=1;c<=nc;c++)
{
if(r%2==0)printf("%2c",L); else printf("%2c",U);
}
printf("\n"); if(r%2==0)L++;else U++;
}
getch();
}
```

Enter no of rows and columns 10 20

```
A A A A A A A A A A A A A A A A A A
a a a a a a a a a a a a a a a a a
B B B B B B B B B B B B B B B B B
b b b b b b b b b b b b b b b b b
C C C C C C C C C C C C C C C C C
c c c c c c c c c c c c c c c c c
D D D D D D D D D D D D D D D D D
d d d d d d d d d d d d d d d d d
E E E E E E E E E E E E E E E E E
e e e e e e e e e e e e e e e e e
```

TC

```
char U='A', L='a';
for(r=1; r<=nr; r++)
{
```

```
  for ( c=1; c<=nc; c++)
  {
```

```
    if(✓%2==0) p( L ); else p( U );
```

```
  }
```

```
  if(r%2==0) L++; else U++;
  p("\n");
}
```

nr nc r c

4 4 1✓ 1 2 3 4

2✓ 1 2 3 4

3

U L r

A a 1

B b 2

3

1 - A A A

AAAA ✓
aaaa ✓
BBBB ✓
bbbb

[illegible]

```

for(r=1;r<=nr; r++)
{
    char U='A', L='a';
    for ( c=1; c<=nc; c++)
    {
        if(c%2==0) p(L++); else p(U++);
    }
    if(r%2==0) L++; else U++;
    p("\n");
}

```

$\frac{nr}{4}$ $\frac{nc}{4}$ $\frac{r}{1}$ $\frac{c}{1\ 2\ 3\ 4}$

2 ✓ 1 2 3 4
 3

AaBbCcDd
 AaBbCcDd
 AaBbCcDd
 AaBbCcDd
 1 3 5

A	a	B	b	c	c
1	2	3	4	5	6

```
TC
File Edit Run Compile Project Options Debug Break/watch
Line 13 Col 53 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int nr, nc,r,c;
clrscr();
printf("Enter no of rows and columns "); scanf("%d %d",&nr, &nc);
for(r=1;r<=nr;r++)
{
char ch=64+nc;
for(c=1;c<=nc;c++)
{
if(r%2==0)printf("%2c",ch--); else printf("%2c",64+c);
}
printf("\n");
}
getch();
}
```

Enter no of rows and columns 10 26

```
A B C D E F G H I J K L M N O P Q R S T U V W X Y Z
Z Y X W V U T S R Q P O N M L K J I H G F E D C B A
A B C D E F G H I J K L M N O P Q R S T U V W X Y Z
Z Y X W V U T S R Q P O N M L K J I H G F E D C B A
A B C D E F G H I J K L M N O P Q R S T U V W X Y Z
Z Y X W V U T S R Q P O N M L K J I H G F E D C B A
A B C D E F G H I J K L M N O P Q R S T U V W X Y Z
Z Y X W V U T S R Q P O N M L K J I H G F E D C B A
A B C D E F G H I J K L M N O P Q R S T U V W X Y Z
Z Y X W V U T S R Q P O N M L K J I H G F E D C B A
```

}

```
TC
File Edit Run Compile Project Options Debug Break/watch
Line 12 Col 66 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int nr,r,c;
clrscr();
printf("Enter no of rows "); scanf("%d",&nr);
for(r=1;r<=nr;r++)
{
for(c=1;c<=nr;c++)
{
if(c==r)printf("1 "); else if(r>c)printf("2 ");else printf("0 ");
}
printf("\n");
}
getch();
}
```

Enter no of rows 3

```
1 0 0
2 1 0
2 2 1
```



```
TC
Enter no of rows 10
1 0 0 0 0 0 0 0 0 0
2 1 0 0 0 0 0 0 0 0
2 2 1 0 0 0 0 0 0 0
2 2 2 1 0 0 0 0 0 0
2 2 2 2 1 0 0 0 0 0
2 2 2 2 2 1 0 0 0 0
2 2 2 2 2 2 1 0 0 0
2 2 2 2 2 2 2 1 0 0
2 2 2 2 2 2 2 2 1 0
2 2 2 2 2 2 2 2 2 1
```

```
if( r==c ) p(1);
else if(r>c) p(2);
else p(0);
```

1	0	0
1,1	1,2	1,3
2	1	0
2,1	2,2	2,3
2	2	1
3,1	3,2	3,3


```
TC
File Edit Run Compile Project Options Debug Break/watch
Line 14 Col 40 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int nr,r,c;
clrscr();
printf("Enter no of rows "); scanf("%d",&nr);
for(r=1;r<=nr;r++)
{
for(c=1;c<=nr;c++)
{
if(c==r){textcolor(LIGHTRED);cprintf("1 ");}
else if(r>c){textcolor(LIGHTGREEN);cprintf("2 ");}
else {textcolor(YELLOW);cprintf("0 ");}_
}
printf("\n");
}
getch();
}
```

Enter no of rows 10

```
1 0 0 0 0 0 0 0 0 0
2 1 0 0 0 0 0 0 0 0
2 2 1 0 0 0 0 0 0 0
2 2 2 1 0 0 0 0 0 0
2 2 2 2 1 0 0 0 0 0
2 2 2 2 2 1 0 0 0 0
2 2 2 2 2 2 1 0 0 0
2 2 2 2 2 2 2 1 0 0
2 2 2 2 2 2 2 2 1 0
2 2 2 2 2 2 2 2 2 1
```

The image shows two screenshots of the Turbo C++ (TC) IDE. The top screenshot displays the source code for a C program that prints a star pattern. The code is as follows:

```
File Edit Run Compile Project Options Debug Break/watch
Line 13 Col 13 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int nr,r,c;
clrscr();
printf("Enter no of rows "); scanf("%d",&nr);
for(r=1;r<=nr;r++)
{
for(c=1;c<=r;c++)
{
printf("*");
}
printf("\n");
}
getch();
}
```

The bottom screenshot shows the execution of the program. The user has entered '4' for the number of rows, and the program has printed the following star pattern:

```
Enter no of rows 4
*
**
***
****
```

The IDE interface includes a menu bar at the top with options like File, Edit, Run, Compile, Project, Options, Debug, and Break/watch. A status bar at the bottom of each window shows the current line and column (Line 13, Col 13) and the file name (E:2PM.C). The Windows taskbar is visible at the bottom of the screen, showing various application icons and the system clock (3:43 PM, 14-Sep-24).

```

TC
File Edit Run Compile Project Options Debug Break/watch
Line 13 Col 13 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int nr,r,c;
clrscr();
printf("Enter no of rows "); scanf("%d",&nr);
for(r=1;r<=nr;r++)
{
for(c=1;c<=r;c++)
{
printf("*");
}
printf("\n");
}
getch();
}

```

for(r=1;r<=4;r++)	<u>nr</u>	<u>r</u>	<u>c=1 to r</u>	
{	4	1	1 to 1 →	1 - ★
for(c=1; c<=r; c++)		2	1 to 2 →	2 - ★ ★-2
{		3	1 to 3 →	3 - ★ ★ ★-3
p(*);		4	1 to 4 →	4 - ★ ★ ★ ★-4
}				
p("\n");				
}				

Home work:

nr=4

```

1
2 3
4 5 6
7 8 9 10

```

nr=4

```

10
9 8
7 6 5
4 3 2 1

```

nr=3

```

6
5 4
3 2 1

```

1
2 5
3 6 8
4 7 9 10